

Recursive trees grown from dynamic building sequences

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Abstract

We study the depth D_n of node n in a preferential recursive tree grown under dynamic affinities with age index $\Lambda \in [0, 2]$. Exact and asymptotic formulae for the mean and the variance are derived. As a consequence, $D_n/\ln n$ converges in probability to $2/(1 + \Lambda)$. We then establish the central limit theorem

$$\frac{D_n - \mathbb{E}[D_n]}{\sqrt{\ln n}} \xrightarrow{L} \mathcal{N}(0, V_\Lambda), \quad V_\Lambda = \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3}.$$

The proof of asymptotic normality proceeds from the parent-index decomposition of the centred depth, an averaging lemma for the attachment kernel, and convergence of all moments via a mixture recurrence. Numerical Monte Carlo experiments for the classical age indices $\Lambda \in \{0, 1, 2\}$ illustrate the mean/variance asymptotics, the weak law, the parent-index limit law and the CLT.

Keywords: random recursive tree; preferential attachment; dynamic affinities; depth; central limit theorem.

1 Introduction

The recursive tree is a classical random structure. Early applications appear in the analysis of pyramid schemes ([Gastwirth and Bhattacharya, 1984](#)) and in philology ([Najock and Heyde, 1982](#)).

Starting with one node at time 0, at each new tick of discrete time, a new node appears and is adjoined by an edge to an existing tree node, which we call the parent. Each node is labeled with its order of appearance in the tree. The operation of letting a parent acquire a new node (child) is often viewed as recruiting.

So, the initial node is the first in the structure and takes the label 1, the second node to appear takes the label 2, and so on. Generally, the n th appearance gives the node generated at that point the label n . Note the unit shift between the time of appearance and the label given to the node born at this time.

When the node labeled n joins the tree, there are $n - 1$ nodes, any of which may become the parent. So, we consider a probability distribution on the set $\{1, \dots, n - 1\}$ to parent the node labeled n , and each such distribution corresponds to an insertion algorithm. No matter what insertion algorithm (emerging from a specific probability model) is used, it induces a property along the labels on root-to-leaf paths: the labels on such a path are in increasing order.

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Therefore, several authors call these structures “increasing trees” (see [Bergeron et al. \(1992\)](#), for example).

Nakata and Mahmoud ([Nakata and Mahmoud, 2024](#)) consider a rather general probability model, in which a static arbitrary sequence $\{a_k\}_{k=1}^{\infty}$ of positive real numbers is determined. They call $\{a_k\}_{k=1}^{\infty}$ the affinity sequence, and consider a_k the affinity of node k after its appearance in the tree. For $n \geq 2$, let $A_{n,k}$ be the event that node k is the chosen parent for node n (when the tree size is $n - 1$). The affinities are the driving force behind the probability model; affinities are turned into probabilities by dividing by $s_{n-1} := \sum_{k=1}^{n-1} a_k$. In other words, the probability model is

$$\mathbb{P}(A_{n,k}) = \frac{a_k}{s_{n-1}}, \quad k = 1, \dots, n-1.$$

This probability model represents an entire class of recursive tree insertion algorithms with general affinities. It is a static preferential model with a_k specified ahead of time and remaining the same throughout the life of the tree after time $k - 1$; a node recruits according to a probability proportional to its affinity.

While the model in ([Nakata and Mahmoud, 2024](#)) is broad and captures many known special cases, the accelerating innovation in technology is dictating the need for additional growth models. The model in ([Nakata and Mahmoud, 2024](#)) covers the standard (uniform) recursive tree ([Drmotá, 2009](#); [Frieze and Karoński, 2016](#); [Hofri and Mahmoud, 2018](#); [Smythe and Mahmoud, 1995](#)), the young-age recruiting tree ([Lyon and Mahmoud, 2020](#)) and the power-weight tree ([Lyon and Mahmoud, 2022](#)). However, such a static affinity model does not cover a case in which the affinity of node k is a function of both k and n . There is need to consider affinities of the form $a_{n,k}$. For instance, the old-age model with affinities $a_{n,k} = n - k$ stands for a rather natural application, where the nodes with smaller indices (older in the tree) have higher recruiting probabilities. Such a dynamic affinity takes “experience” into account.

To be specific, in this manuscript we study the recursive tree with the dynamic affinities

$$a_{n,k} = \alpha n + \beta k > 0, \quad k = 1, \dots, n-1, \quad (1)$$

subject to $2\alpha + \beta > 0$. Writing $s_{n-1} = \sum_{k=1}^{n-1} a_{n,k}$, the attachment probabilities are

$$\mathbb{P}(A_{n,k}) = \frac{a_{n,k}}{s_{n-1}}, \quad k = 1, \dots, n-1.$$

The hypothesis $2\alpha + \beta > 0$ ensures that s_{n-1} is positive and strictly increasing in n .

It is convenient to reduce the two parameters to a single age index

$$\Lambda = \frac{2\alpha}{2\alpha + \beta},$$

which ranges over the interval $[0, 2]$. In these coordinates, the event $A_{n,k}$ (node k recruits node n) has probability

$$\mathbb{P}(A_{n,k}) = \frac{\alpha n + \beta k}{(2\alpha + \beta) n(n-1)/2} = \frac{\Lambda n + 2(1 - \Lambda)k}{n(n-1)}, \quad k = 1, \dots, n-1.$$

Remark 1. The dynamic affinities in (1) contain the following models:

- **Young-age model.** Take $\alpha = 0$ and $\beta > 0$ (hence $\Lambda = 0$). The affinities are $a_{n,k} = \beta k$, and

$$\mathbb{P}(A_{n,k}) = \frac{2k}{n(n-1)}, \quad k = 1, \dots, n-1.$$

This model appears in ([Lyon and Mahmoud, 2020, 2022](#)). Nodes with high index k are still young.

- **Uniform model.** Take $\alpha > 0$ and $\beta = 0$ (hence $\Lambda = 1$). The affinities are $a_{n,k} = \alpha n$, and

$$\mathbb{P}(A_{n,k}) = \frac{1}{n-1}, \quad k = 1, \dots, n-1.$$

This classical model has been studied extensively; see (Drmota, 2009; Frieze and Karoński, 2016; Smythe and Mahmoud, 1995).

- **Old-age model.** Take $\alpha > 0$ and $\beta = -\alpha$ (hence $\Lambda = 2$). The affinities are $a_{n,k} = \alpha(n-k)$, and

$$\mathbb{P}(A_{n,k}) = \frac{2(n-k)}{n(n-1)}, \quad k = 1, \dots, n-1.$$

This model appears in (Hofri and Mahmoud, 2018). Early nodes have high affinity.

Figure 1 depicts the six recursive old-age preferential trees of size 4 together with their probabilities.

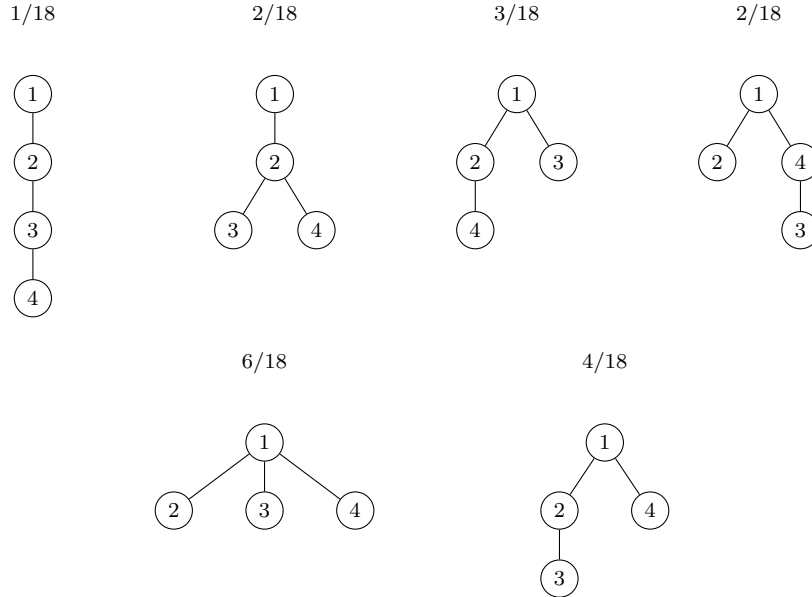


Figure 1: Old-age preferential recursive trees of size 4 and illustrative probabilities (affinity of node k proportional to $n - k$).

2 Notation

The indicator of an event E is denoted by $\mathbb{I}_E$. We use Iverson's brackets: for a predicate B , the symbol $\llbracket B \rrbracket$ equals 1 if B is true and 0 otherwise. For any expression g , we adopt the convention that $\llbracket B \rrbracket g = 0$ whenever B is false, even if g is formally undefined or infinite.

Several results are expressed in terms of harmonic numbers. For an integer $r \geq 1$, the sum $\sum_{k=1}^n 1/k^r := H_n^{(r)}$ is the n th harmonic number of order r . It is customary to drop the superscript for the first-order harmonic number and to write H_n in place of $H_n^{(1)}$. The harmonic numbers admit the classical expansion

$$H_n = \ln n + \gamma + O\left(\frac{1}{n}\right), \quad (2)$$

where $\gamma = 0.57721 \dots$ is the Euler–Mascheroni constant.

The symbol $\xrightarrow{\text{P}}$ denotes convergence in probability, and $\xrightarrow{\text{L}}$ denotes convergence in distribution.

3 Depth

Let D_n be the depth of node n in a recursive tree grown under preferential dynamic affinities. In this section we derive exact and asymptotic formulae for the mean and the variance of D_n . The resulting growth rates yield a weak law of large numbers by Chebyshev's inequality. The limiting distribution is established in Subsection 3.3.

Let $\psi_n(t) = \mathbb{E}[e^{D_n t}]$ be the moment generating function of D_n . For $n \geq 2$, the depth satisfies the recursion

$$D_n = \sum_{k=1}^{n-1} (1 + D_k) \mathbf{I}_{A_{n,k}}. \quad (3)$$

In the present model the affinities depend only on the labels, so $A_{n,k}$ is independent of D_k . Consequently, the sequence $\psi_n(t)$ obeys

$$\psi_n(t) = \frac{e^t}{n(n-1)} \sum_{k=1}^{n-1} (\Lambda n + 2(1-\Lambda)k) \psi_k(t). \quad (4)$$

It is convenient to write $n(n-1) = 2\binom{n}{2}$; we adopt this abbreviation throughout.

The recurrence (4) requires two successive differencing steps in order to eliminate the cumulative sums. This is the principal technical difference with the static-affinity models treated in (Nakata and Mahmoud, 2024), which typically require only one differencing step.

After two differencing steps and simplification, one obtains, for $n \geq 3$,

$$\begin{aligned} \binom{n}{2} \psi_n(t) - \left(2\binom{n-1}{2} + e^t \left(\left(1 - \frac{\Lambda}{2}\right)n - (1-\Lambda) \right) \right) \psi_{n-1}(t) \\ + \left(\binom{n-2}{2} + e^t \left(1 - \frac{\Lambda}{2}\right)(n-2) \right) \psi_{n-2}(t) = 0. \end{aligned} \quad (5)$$

3.1 The mean depth

Taking derivatives of (5) with respect to t and evaluating at $t = 0$, for all $n \geq 2$, we get the recurrence

$$\begin{aligned} \binom{n}{2} \mathbb{E}[D_n] - \left\{ 2\binom{n-1}{2} + \left(1 - \frac{\Lambda}{2}\right)n - (1-\Lambda) \right\} \mathbb{E}[D_{n-1}] \\ + \left\{ \binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right)(n-2) \right\} \mathbb{E}[D_{n-2}] = 1. \end{aligned} \quad (6)$$

In the solution to this recurrence, certain sums of ratios of gamma functions appear. Starting from

$$\frac{\Gamma(i)}{\Gamma(i-\Lambda)} = \begin{cases} (i-1)(i-2), & \text{if } \Lambda = 2, \\ \frac{1}{\Lambda+1} \left(\frac{\Gamma(i+1)}{\Gamma(i-\Lambda)} - \frac{\Gamma(i)}{\Gamma(i-\Lambda-1)} \right), & \text{if } \Lambda \neq 2, \end{cases}$$

the sum on the index $i = 3, \dots, n$ is telescopic, yielding

$$\sum_{i=3}^n \frac{\Gamma(i)}{\Gamma(i-\Lambda)} = \frac{\Gamma(n+1)}{(\Lambda+1)\Gamma(n-\Lambda)} - \mathbb{I}[\Lambda \neq 2] \frac{2}{(\Lambda+1)\Gamma(2-\Lambda)}. \quad (7)$$

Proposition 1. *Let D_n be the depth of node n in a preferential recursive tree of age index Λ . For $n \geq 2$,*

$$\begin{aligned}\mathbb{E}[D_n] &= \frac{2}{\Lambda+1}H_{n-1} + \frac{\Lambda-1}{\Lambda+1} \\ &\quad + \llbracket \Lambda \neq 2 \rrbracket \frac{2(\Lambda-1)}{(\Lambda+1)\Gamma(2-\Lambda)} \sum_{i=3}^n \frac{\Gamma(i-\Lambda)}{(i-1)\Gamma(i+1)}.\end{aligned}$$

Proof sketch. Using the identity

$$2\binom{n-1}{2} + \left(1 - \frac{\Lambda}{2}\right)n - (1 - \Lambda) = \binom{n}{2} + \left(\binom{n-2}{2} + \left(1 - \frac{\Lambda}{2}\right)(n-2)\right), \quad (8)$$

rewrite (6) in incremental form. Put $a_n = \mathbb{E}[D_n] - \mathbb{E}[D_{n-1}]$. Then for $n \geq 2$,

$$a_n = \frac{(n-2)(n-1-\Lambda)}{n(n-1)}a_{n-1} + \frac{2}{n(n-1)},$$

with $a_2 = 1$. Solving this linear recurrence and applying (7) yields

$$a_n = \frac{2}{(n-1)(\Lambda+1)} + \llbracket \Lambda \neq 2 \rrbracket \frac{2(\Lambda-1)\Gamma(n-\Lambda)}{(\Lambda+1)\Gamma(2-\Lambda)(n-1)\Gamma(n+1)}.$$

Unwinding a_n from $\mathbb{E}[D_2] = 1$ gives the stated formula. □

Corollary 1 (Young-age). *For $\Lambda = 0$ and $n \geq 3$, $\mathbb{E}[D_n] = 2(H_n - 1)$.*

Corollary 2 (Uniform). *For $\Lambda = 1$ and $n \geq 1$, $\mathbb{E}[D_n] = H_{n-1}$.*

Corollary 3 (Old-age). *For $\Lambda = 2$ and $n \geq 2$, $\mathbb{E}[D_n] = \frac{2}{3}H_{n-1} + \frac{1}{3}$.*

Lemma 1. *Let $\Lambda \in [0, 2]$. Then for $i \geq 3$,*

$$\frac{\Gamma(i-\Lambda)}{\Gamma(i+1)} \leq \frac{3^\Lambda}{2i^{\Lambda+1}\Gamma(3-\Lambda)}.$$

Proposition 2. *As $n \rightarrow \infty$,*

$$\mathbb{E}[D_n] = \frac{2}{1+\Lambda}H_{n-1} + C_\Lambda + O(n^{-(\Lambda+1)}),$$

where

$$C_\Lambda = \frac{\Lambda-1}{\Lambda+1} + \llbracket \Lambda \neq 2 \rrbracket \frac{2(\Lambda-1)}{(\Lambda+1)\Gamma(2-\Lambda)} \sum_{i=3}^{\infty} \frac{\Gamma(i-\Lambda)}{(i-1)\Gamma(i+1)}.$$

3.2 The variance of the depth

Proposition 3. *We have $\text{Var}[D_1] = \text{Var}[D_2] = 0$, and for $n \geq 3$ an exact (though lengthy) expression for $\text{Var}[D_n]$ follows by solving the second-moment recurrence obtained from (5); see the companion computational notebook for the expanded form. For the classical integer indices one obtains closed forms.*

Corollary 4 (Young-age variance). *For $\Lambda = 0$, $\text{Var}[D_n] = 2H_n - 4H_n^{(2)} + 2$.*

Corollary 5 (Uniform variance). *For $\Lambda = 1$, $\text{Var}[D_n] = H_{n-1} - H_{n-1}^{(2)}$.*

Corollary 6 (Old-age variance). *For $\Lambda = 2$ and $n \geq 3$,*

$$\text{Var}[D_n] = \frac{10}{27}H_{n-1} + \frac{12}{27}H_n^{(2)} - \frac{28}{27} + \frac{12}{27n}.$$

Corollary 7. *As $n \rightarrow \infty$,*

$$\text{Var}[D_n] = \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3}H_{n-1} + O(1).$$

Since $\text{Var}[D_n] = O(\ln n)$ while $\mathbb{E}[D_n] \sim a \ln n$ with $a = 2/(1 + \Lambda)$, Chebyshev's inequality yields the weak law.

Corollary 8. *As $n \rightarrow \infty$,*

$$\frac{D_n}{\ln n} \xrightarrow{\text{P}} \frac{2}{\Lambda + 1}.$$

3.3 Asymptotic distribution

Set

$$\mu_n := \mathbb{E}[D_n], \quad \Delta_n := D_n - \mu_n, \quad a := \frac{2}{1 + \Lambda}, \quad V_\Lambda := \frac{2 + 8\Lambda - 2\Lambda^2}{(\Lambda + 1)^3},$$

so that $\text{Var}[D_n] = V_\Lambda H_{n-1} + O(1) \sim V_\Lambda \ln n$.

Theorem 1. *Let $\Lambda \in [0, 2]$. As $n \rightarrow \infty$,*

$$\frac{D_n - \mu_n}{\sqrt{\ln n}} \xrightarrow{\text{L}} \mathcal{N}(0, V_\Lambda).$$

Equivalently,

$$\frac{D_n - \mu_n}{\sqrt{\text{Var}[D_n]}} \xrightarrow{\text{L}} \mathcal{N}(0, 1).$$

The argument is organised as follows. Write $p_{n,k} := \mathbb{P}(A_{n,k})$ and let J_n be the parent of node n . Centring gives the exact decomposition

$$\Delta_n = \Delta_{J_n} + (\mu_{J_n} - \mathbb{E}[\mu_{J_n}]) = \Delta_n^{(1)} + \Delta_n^{(2)}. \quad (9)$$

Lemma 2 identifies the limiting law of J_n/n . Lemma 3 shows that the mean-fluctuation term $\Delta_n^{(2)}$ is negligible on the scale $\sqrt{\ln n}$. An averaging lemma for the attachment kernel then yields, by induction, convergence of all moments of $\Delta_n/\sqrt{\ln n}$ to those of $\mathcal{N}(0, V_\Lambda)$. The CLT follows from the Fréchet–Shohat theorem.

Lemma 2. *As $n \rightarrow \infty$, $J_n/n \xrightarrow{\text{L}} J$, where J is supported on $(0, 1)$ with density*

$$f_\Lambda(x) = \Lambda + 2(1 - \Lambda)x, \quad x \in (0, 1). \quad (10)$$

In particular, $\mathbb{P}(J > 0) = 1$ and $\mathbb{E}[|\ln J|^q] < \infty$ for every $q \geq 1$.

Lemma 3. *As $n \rightarrow \infty$,*

$$\Delta_n^{(2)} \xrightarrow{\text{L}} W := a \ln J - a \mathbb{E}[\ln J].$$

Consequently $\Delta_n^{(2)}/\sqrt{\ln n} \xrightarrow{\text{P}} 0$, all moments of $\Delta_n^{(2)}$ remain bounded, and

$$\mathbb{E}[W^2] = -V_\Lambda \mathbb{E}[\ln J].$$

Lemma 4 (Averaging). *Let $(u_n)_{n \geq 1}$ satisfy $u_1 = 0$ and*

$$u_n = \sum_{k=1}^{n-1} p_{n,k} u_k + f_n, \quad n \geq 2.$$

1. If $|f_n| \leq F(1 + \ln n)^\beta$, then $|u_n| \leq C_F(1 + \ln n)^{\beta+1}$ for some $C_F = C_F(\Lambda, \beta) < \infty$.
2. If $f_n = o((1 + \ln n)^\beta)$, then $u_n = o((1 + \ln n)^{\beta+1})$.

Lemma 5 (Moment convergence). *For every integer $m \geq 1$,*

$$\mathbb{E}[\Delta_n^m] = \mu_m(\ln n)^{m/2} + o((\ln n)^{m/2}),$$

where $\mu_m = \mathbb{E}[Z^m]$ for $Z \sim \mathcal{N}(0, V_\Lambda)$. If m is odd, then moreover $\mathbb{E}[\Delta_n^m] = O((\ln n)^{(m-1)/2})$.

Proof of Theorem 1. Lemma 5 and Carleman's criterion imply that the moments determine the Gaussian law. Fréchet–Shohat yields $\Delta_n/\sqrt{\ln n} \xrightarrow{L} \mathcal{N}(0, V_\Lambda)$. Since $\text{Var}[D_n]/\ln n \rightarrow V_\Lambda$, Slutsky's lemma gives the standardised form as well. $\square$

Remark 2. For $\Lambda \in \{0, 1, 2\}$ one recovers $V_0 = 2$, $V_1 = 1$ and $V_2 = 10/27$, in agreement with the exact variance corollaries. In particular, in the uniform case $\Lambda = 1$, $(D_n - H_{n-1})/\sqrt{\ln n} \xrightarrow{L} \mathcal{N}(0, 1)$.

4 Numerical illustrations

We complement the asymptotic theory with Monte Carlo experiments for the three classical age indices $\Lambda \in \{0, 1, 2\}$. Trees are grown by inverse-CDF sampling of the attachment kernel. For each pair (Λ, n) we estimate $\mathbb{E}[D_n]$ and $\text{Var}[D_n]$ from 800 independent replications; CLT diagnostics use $n = 5000$ and 2000 replications. The Python code that generated all figures and tables is distributed with this manuscript (`python/simulate_dynamic_trees.py`).

A frequent source of apparent disagreement between theory and simulation is the confusion between *exact finite- n formulae* and *asymptotic limits*. For example,

$$\frac{\mathbb{E}[D_n]}{\ln n} = \frac{2}{1 + \Lambda} \cdot \frac{H_{n-1}}{\ln n} + \frac{C_\Lambda}{\ln n} + o\left(\frac{1}{\ln n}\right)$$

converges slowly to $2/(1 + \Lambda)$; likewise $\text{Var}[D_n]/\ln n$ approaches V_Λ only at rate $O(1/\ln n)$. Accordingly, the plots below overlay simulation against the exact corollaries whenever available, and reserve dotted horizontal lines for the asymptotic targets.

Figure 2 displays the theoretical CLT variance coefficient V_Λ on $[0, 2]$. Figure 3 compares empirical means and variances with the exact formulae of Corollaries 3.1–3.6: the match is already close for n of a few hundred. Figure 4 checks the weak law of Corollary 8 at finite n (solid/dashed) and shows the asymptotic limits as dotted references. Figure 5 makes the same point for the variance coefficient $\text{Var}[D_n]/H_{n-1}$.

Parent-index histograms at $n = 5000$ (Figure 6) overlay the limiting density (10). Finally, Figure 7 shows studentized histograms of $(D_n - \mu_n)/\sqrt{\text{Var}[D_n]}$ against $\mathcal{N}(0, 1)$ together with normal QQ-plots. Depths are integer-valued, so the studentized sample lives on a lattice of spacing $1/\sqrt{\text{Var}[D_n]}$. The histograms therefore use lattice-aligned bins of that width; otherwise narrow fixed binning artificially inflates bar heights between atoms and creates a spurious visual conflict with the Gaussian density. The QQ-plots retain the expected staircase from discreteness, while following the line $y = x$. This is the finite- n form of Theorem 1.

Table 1 summarises selected moment estimates. Across all three models the empirical means track the exact formulae within Monte Carlo error, and the empirical variances are consistent with the predicted growth of order $\ln n$.

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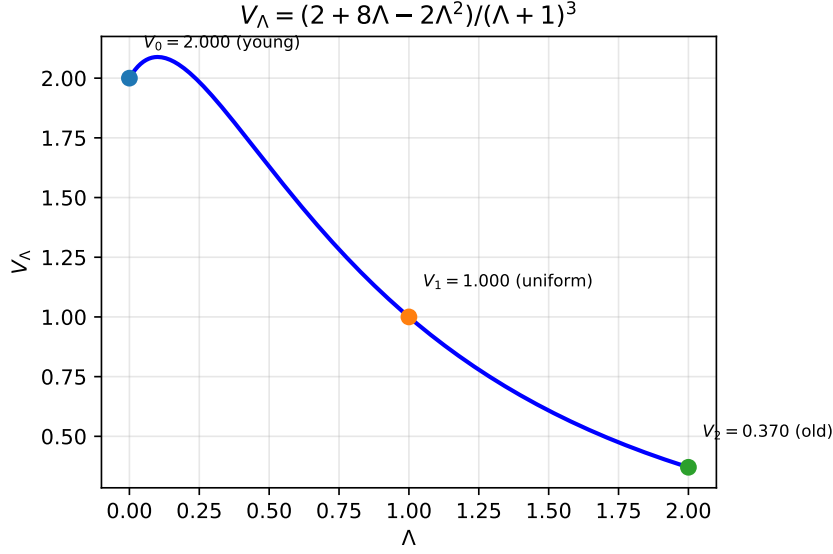


Figure 2: CLT variance coefficient $V_\Lambda = (2 + 8\Lambda - 2\Lambda^2)/(\Lambda + 1)^3$.

Table 1: Selected Monte Carlo checks (800 replications).

Λ	n	emp. mean	exact mean	emp. var	exact var
0	5000	15.999	16.189	13.541	13.610
1	5000	9.173	9.094	7.662	7.450
2	5000	6.434	6.396	3.132	3.062

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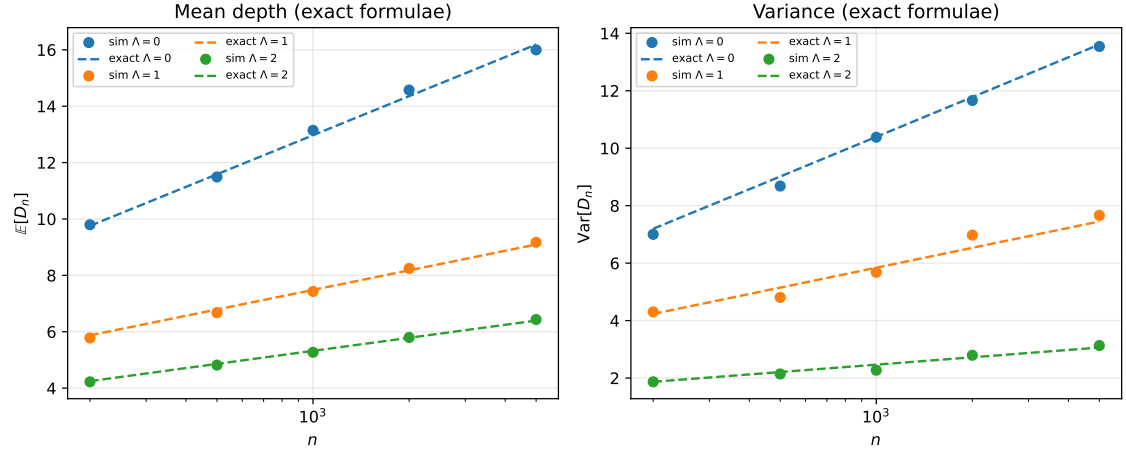


Figure 3: Empirical versus exact mean and variance of D_n for $\Lambda \in \{0, 1, 2\}$ (same colour: markers = simulation, dashed = exact).

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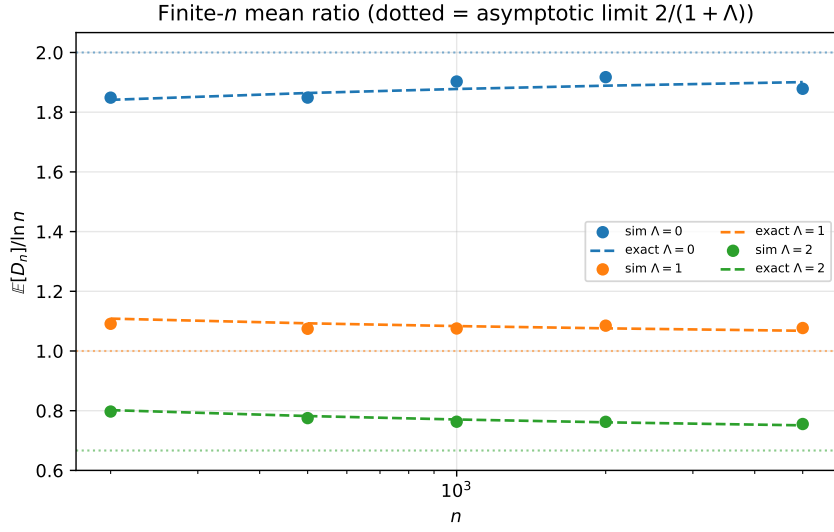


Figure 4: Finite- n check of $\mathbb{E}[D_n]/\ln n$: simulation versus exact mean ratio; dotted lines are the asymptotic limits $2/(1 + \Lambda)$.

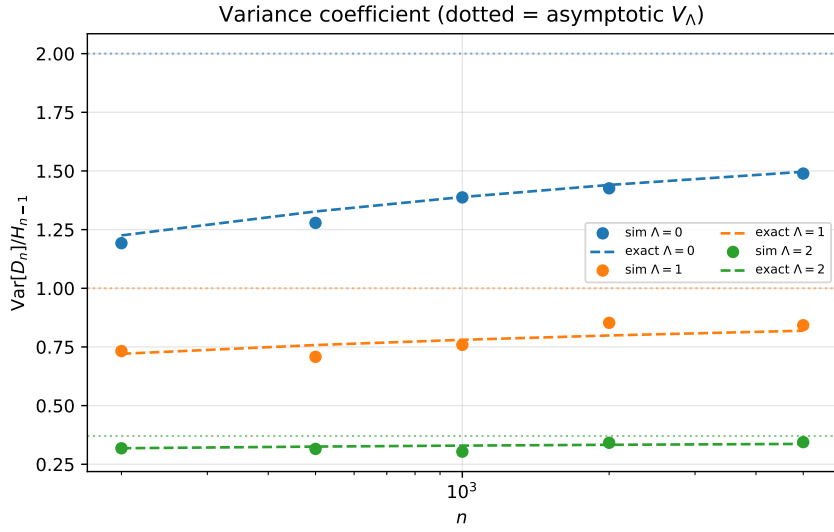


Figure 5: Finite- n variance coefficient $\text{Var}[D_n]/H_{n-1}$ versus the asymptotic target V_Λ (dotted).

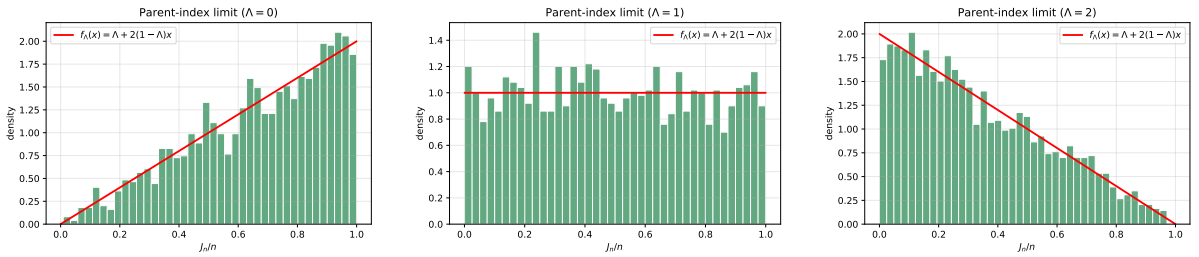


Figure 6: Empirical density of J_n/n versus f_Λ for young-age ($\Lambda = 0$), uniform ($\Lambda = 1$) and old-age ($\Lambda = 2$) trees.

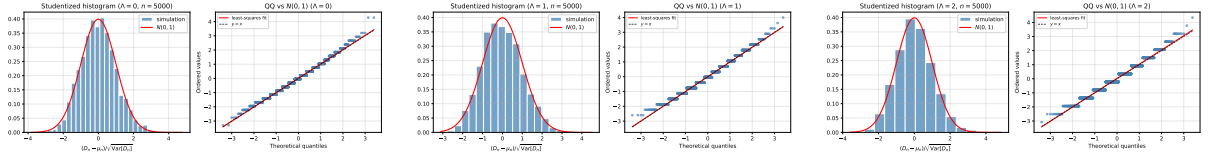


Figure 7: CLT diagnostics at $n = 5000$: lattice-binned studentized histograms of $(D_n - \mu_n)/\sqrt{\text{Var}[D_n]}$ (bin width = $1/\sqrt{\text{Var}[D_n]}$) and QQ-plots against $\mathcal{N}(0, 1)$.